Disc: a haptic display prototype

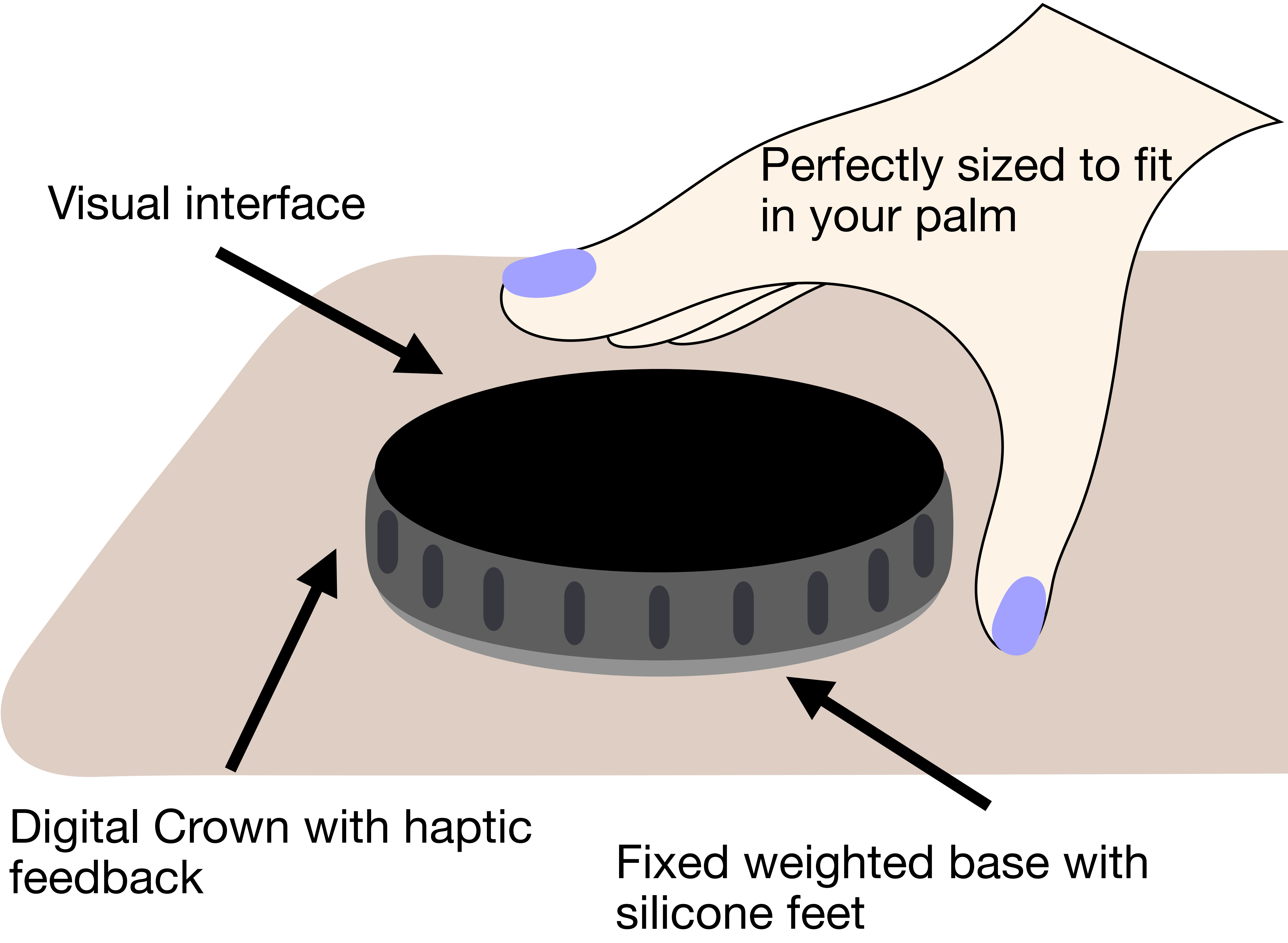
A companion for surfaces,
Keeping you in touch with your everyday activities

Raaghav Thirumaligai

What?

A haptic display you need to touch

- The vertical flutes on stainless steel casing has a surface your fingertips settle into
- Interaction is done with a satisfying buzz and click from the haptic motors
- Rotating the display to interact maintains content orientation by digitally rotating the content to the initial position



Why?

Taps/Swipes < Buzz/Click

- Using your iPhone to start a timer takes
    - tap, unlock, swipe, tap, tap, tap, swipe, tap
    - OR Siri - “start a timer”
- Using your watch does the same thing
- If this lived on your counter it would take a word: “timer” a twist and a click

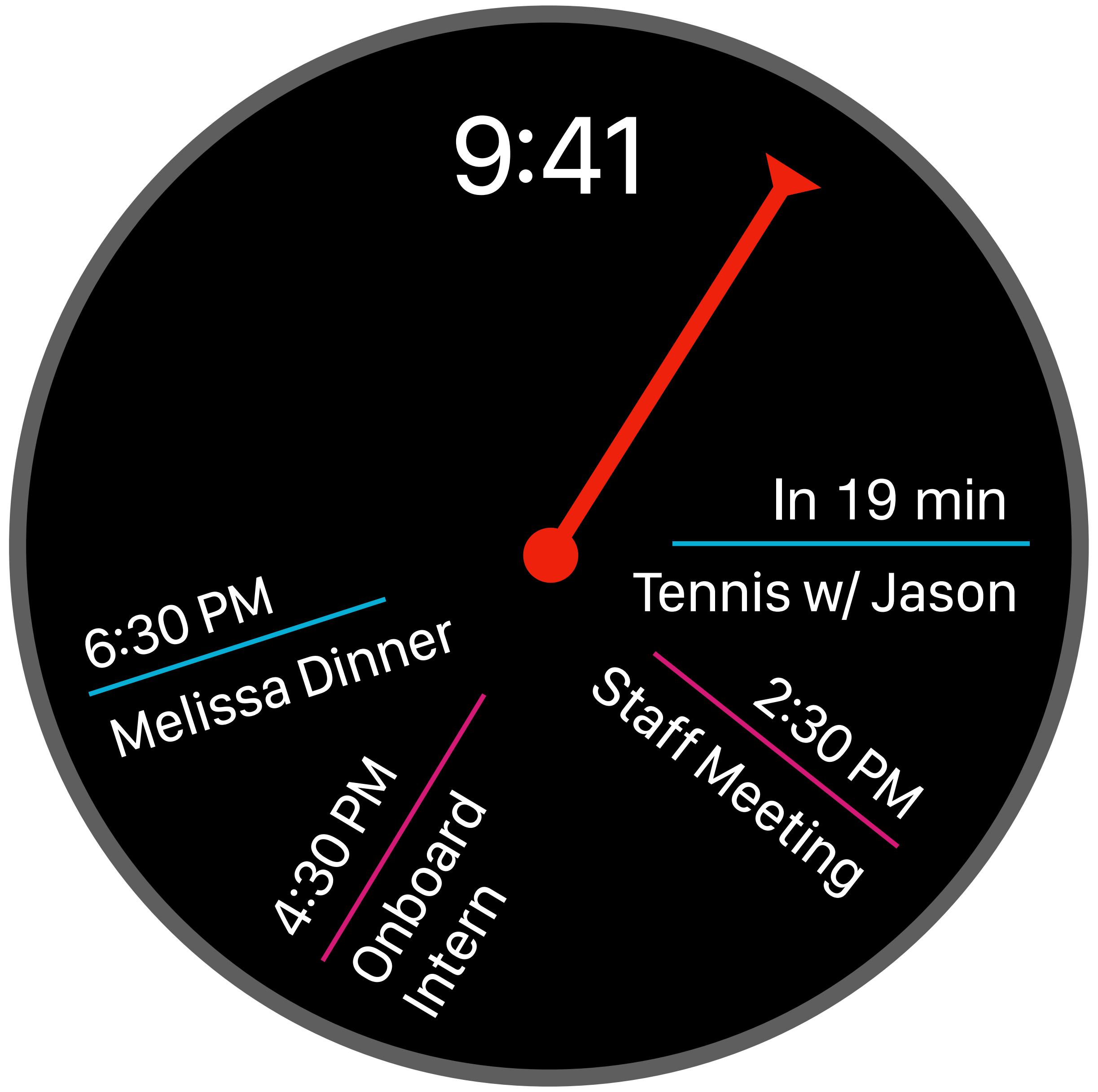
Rotating fast adjusts time quickly

Software decides if we want to skip quickly from high velocity or click slowly from lower velocity

Why… else?

More potential UI-deas

Of course Siri AI but as a conversational control input



New calendar visualization

Give a display to your HomePods or AirPods

How?

First prototype

- Display can be a small LCD
- Quick PLA 3D print a case
- Small DC motor is back driven measuring angular velocity
- Small vibration motor to test “clicks”
- Motor rests on button for “press”
- TPU/Cardboard base for non-slip on surface to rotate against

Potentiometer isn’t needed because we only need relative position

If there isn’t enough resistance, the device will just slip